Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_02

1. Summary of Key Deviations

The analysis of Run_02 shows a clear pattern of fuel over-delivery originating from the pump, with all other systems appearing to react in a physically consistent manner.

- **Flow_Pump (Moderate):** The primary physical anomaly is a significant positive **offset** on the `Flow_Pump` signal, which consistently reads **+12%** higher than the healthy baseline across active states.
- **Flow_Bypass (Moderate):** As a direct consequence of the increased pump flow, the `Flow_Bypass` signal shows an even larger positive **offset**, peaking at **+21.9%**. The system is diverting the excess flow that it cannot send to the engine.
- **Flow_Engine1-4 (Minor):** All four engine flows show a minor positive **offset** (ranging from **+3% to +7%**), indicating that some of the excess fuel is making it to the combustion chamber.
- **System Pressures (Nominal/Minor):** Most system pressures remain nominal. `p_FMU` and `p_Shut` show a very slight **negative offset** in the final states, which is unusual given the high flow but suggests the system is managing to keep pressures stable.
- **Flow_PRV (Nominal):** There is **no significant deviation** on the `Flow_PRV` signal, indicating the system over-performance was not severe enough to trigger the high-pressure relief valve.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Main Fuel Pump**	**Significant Over-performance / Degradation**	This is the most compelling hypothesis. The pump delivering **+12%** more flow than commanded is the single upstream fault that explains all other downstream effects: the excess flow is diverted (**Flow_Bypass** is high) and also forced through to the engines (**Flow_Engine** is high). This could be caused by internal wear or a control-side fault affecting the pump's displacement.

2	**Bypass Valve**	**Stuck Partially Closed / High Cracking Pressure**	A restricted bypass valve would fail to divert the correct amount of flow, forcing more fuel downstream. This would increase `Flow_Engine` and also cause a rise in upstream pressure, forcing the `Flow_Pump` reading up. While plausible, the magnitude of the `Flow_Pump` deviation makes it a more likely source than a faulty valve.
3	**Fuel Metering Unit (FMU)**	**Stuck Partially Open / Over-delivering**	If the FMU were stuck open, it would lead to higher `Flow_Engine` readings. However, this wouldn't, by itself, cause the `Flow_Pump` to read high; in fact, the reduced resistance might even cause pump flow to drop. Therefore, this is unlikely to be the root cause.
4	**Flow_Pump Sensor**	**Calibration Shift (Positive Offset)**	It's possible the sensor is simply reading high. However, the fact that both `Flow_Bypass` and `Flow_Engine` also read high, in a physically consistent manner, strongly suggests there is a real, physical increase in flow originating from the pump. A sensor-only fault is less likely.

5	**System-wide Sensor Calibration Issue**	**Multiple Positive Offsets**	It is a remote possibility that three independent flow sensors (`Flow_Pump`, `Flow_Bypass`, `Flow_Engine`) all have a simultaneous positive calibration error. This is far less likely than a single upstream physical fault (the pump) causing all three to react.

3. Most Probable Hypothesis

The most probable root cause is a **Significant Over-performance of the Main Fuel Pump.**

Reasoning: The evidence strongly points to the pump as the origin of the anomaly. The `Flow_Pump` signal shows a large and persistent positive offset (+12%), indicating a real physical over-delivery of fuel. This single upstream event provides a complete and sufficient explanation for all the other observed deviations: the excess flow is correctly handled by the system, with some being diverted by the bypass valve (`Flow_Bypass` is high) and the rest being sent to the engine (`Flow_Engine` is high). The stability of the pressure readings suggests the control system is successfully managing the excess flow without triggering a dangerous over-pressurization event.